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1. State whether or not a cohort study is best suited for each of the following scenarios:
 - A. when little is known about a rare exposure
 - B. when little is known about a rare disease
 - C. when the study population will be difficult to follow up
 - D. when you want to learn about multiple effects of an exposure
2. indicate whether the following statements are true or false:
 - A. a retrospective cohort study is more efficient than a prospective cohort study for studying diseases with a long latent and induction period
 - B. cohort studies are the most sensible design for examining many exposures in relation to a single disease
 - C. the ideal comparison group for a cohort study would consist of exactly the same individuals in the exposed group had they not be exposed
 - D. loss to follow-up can be a problem in a cohort study but not an experimental study
3. Which of the following techniques that are commonly used in experimental studies can also be applied to cohort studies?
 - A. Blinding
 - B. Placebo
 - C. randomization
 - D. Run-in period
4. Indicate whether the following statements are true or false:
 - A. it is possible to obtain a valid estimate of disease prevalence from a typical case-control study
 - B. the purpose of a control group in a case-control study is to provide information on the disease distribution in the source population that produced the cases
 - C. the control group in a case-control study should never include individuals who have the cases's disease
 - D. a case-control study is the most efficient design for studying the health effects of rare exposures, while a cohort study is the most efficient design for studying the risk factors for rare diseases
 - E. case identification is generally more difficult than control identification in case-control studies
 - F. the odds of illness are mathematically equivalent to the risk of illness
5. briefly define each of the following terms:
 - A. Recall Bias
 - B. control selection bias
6. Indicate whether the following statements are true or false:
 - A. a study must be valid before its results can be generalized

- B. Bias is introduced primarily during the analysis stage of a study
 - C. bias can pull an estimate of association either toward the null or away from the null
 - D. using an inaccurate case definition increases the likelihood of non-differential misclassification of the disease
 - E. including a large sample size reduces self-selection bias
 - F. poor recall and recall bias are synonymous terms for the same concept
7. Selection bias is most likely to occur in:
- A. case-control studies
 - B. retrospective cohort studies
 - C. experimental studies
 - D. both retrospective and case-control
 - E. both retrospective and experimental
8. Recall bias is most likely to occur in:
- A. case-control studies
 - B. prospective cohort studies
 - C. experimental
 - D. all of the above
 - E. none of the above
9. Indicate whether the following statements are true or false:
- A. all high-quality epidemiological studies include techniques for controlling confounding
 - B. Intermediate variables in a causal pathway are special types of confounders
 - C. The counterfactual ideal is used to guide the selection of a comparison group in order to minimize confounding
 - D. epidemiologists can tell if confounding is present by examining the strength of the crude measure of association
 - E. experimental studies always have less confounding than observational studies
10. Define the following terms:
- A. Chance
 - B. precision
 - C. Statistical inference
11. What is the main assumption involved in hypothesis testing and the calculation of P values?
12. What is the main limitation of significance testing?
13. Why are P values considered confounded statistics?
14. Give one reason why many epidemiologists prefer to use confidence intervals instead of P values to assess the role of random error

15. Indicate whether the following statements are true or false:
- A. Unlike bias and confounding, random errors are unsystematic
 - B. reducing random error also reduces errors from bias and confounding
 - C. precise exposure data can be achieved by repeating the exposure measurements
 - D. Increasing the sample size decreases the change of selecting an unrepresentative sample
 - E. a statistically significant finding always has public health significance
16. Epidemiologists plan the appropriate size for a study by:
- A. using judgement, experience, and intuition
 - B. performing sample size calculations
 - C. both A and B
 - D. Neither A nor B
17. Describe two methods for controlling confounding during the analysis, and give one advantage and one disadvantage for each method
18. Indicate whether the following statements are true or false:
- A. Hill's guidelines of temporality is more easily established in a prospective than retrospective study
 - B. Strong associations are more likely to be causal than weak ones because they are less likely to be due to alternative explanations
 - C. a causal relationship between an exposure and disease cannot be established unless all of Hill's guidelines are met
 - D. Hill's guidelines of specificity means that an exposure can cause only one disease
 - E. exposure to HIV is a necessary cause of AIDS
 - F. Exposure to HIV is a sufficient cause of AIDS
 - G. exposure to cigarette smoke is a necessary cause of lung cancer
 - H. exposure to cigarette smoke is a sufficient cause of lung cancer
19. When do you accept or reject the null hypothesis?
20. When to use a case-control study vs. a cohort
21. What is the 3 criteria for a confounder?
22. How can you control for confounding?
23. What are the three requirements for a screening test?